Exercise 9 - Queues and Vectors

This weeks exercise will focus on implementing the `queue` *data structure from the* `std_library`.
You will also be implementing more `vectors` *from the* `std_library`.

If you have any questions or problems, please write in the Moodle!

Exercise 1 — Checkout at a store - Groundwork

A store nearby wants to model their workforce at a checkout. For that we gonna implement a `Class Cashier`.

For this Exercise write your `Class and Structs` *into* `main.cpp` *as well as trying out your code in* `main.cpp`.

a) Declare the `struct Item` having a field `double price`, which can be read via `getPrice()`, but should not be modifiable after initiation.

b) Declare the `struct Purchase` having the field `std::vector<Item> items`. For the `std::vector items` write a `wrapper method` called `add_item(Item i)`, which adds an item to the end of the `std::vector items`.

Exercise 2 — Checkout at a store - Implementing the Checkout Queue

a) Declare a `Class Cashier` *with a private* `std::queue<Purchase> checkoutCounter` *field.*

b) Write a method `placePurchase(Purchase p)`, *which uses* `push()` *from* `std::queue` *to attach a Purchase onto the* `checkoutCounter`.

c) Finally write a method `collectPurchase` *using* `pop()` *from* `std::queue` *collect the* `Purchase` *by pushing it off the* `checkoutCounter`.